# Catch the Photon

Facilitator guide for numbered wave cards

## Goal and materials

Make your detector bright and the other detector dark, then sample one photon. For two players, allow about 10 minutes. Use the 40 printed wave cards, the Wave Board, two tokens, and blue/purple chips or paper slips. The board can require up to 32 slips of each color. Use identical opaque card backs or sleeves.

## Read a card

The number is wave strength (amplitude): move that many squares in the arrow direction. Blue ↑ is up, red ↓ down, purple → right, gold ← left. Each direction has seven strength-1 and three strength-2 cards. All plots use the same scale and frequency. The arrow describes a board move, not a photon's travel.

## Set up and play eight rounds

Put both detector tokens at START (brightness 0). Shuffle and deal three cards each. Each player secretly chooses one card and a target detector. Reveal both, resolve A's card first and then B's, and draw back to three cards. Apply the full 1- or 2-square move. Keep used cards in a discard pile; reshuffle them if the draw pile runs out.

## Edges and simultaneous moves

A player cannot choose a card that would leave the board on the selected detector. If the first revealed move makes the second move leave the board, the second card is lost. If no card is legal on either detector, discard one instead. A 2-square move cannot be shortened to fit.

## Catch the photon and replay

After eight rounds, read the brightness under each token. Put that many chips of each player's color in a cup and draw one. A brighter detector is more likely to win, but can lose a trial. If both are dark, play another round. In the browser, Solo quick play starts the computer game with open hands and resolves rounds in fewer clicks. New game or Play again keeps that mode.

## An exact cancellation example

At one square above START, a ↓1 card returns the token to START and makes brightness 0. A ↓2 card instead crosses START and ends one square below it: brightness stays 1. A larger card is not always better.

# The science and discussion

## Science and probability

At position (x, y), the model's wave is $y \cos(2\pi t/T) + x \sin(2\pi t/T)$, with amplitude $\sqrt{(x^2 + y^2)}$. The board number $x^2 + y^2$ is brightness: an unnormalized probability weight. Divide each detector's weight by their sum to obtain the detection probabilities. Weights 25 and 9 give chances 25/34 and 9/34. When both weights are zero, the game has no photon draw.

## Phase and strength

Up begins at a crest, down at a trough, right at zero rising, and left at zero falling. In-step amplitudes add. Opposite phases subtract; complete cancellation requires equal amplitudes. ↑2 + ↓1 leaves ↑1, while ↑2 + ↓2 gives zero. Equal-amplitude waves a quarter-cycle apart combine to an amplitude $\sqrt{2}$ times either wave. A strength-2 card from START gives brightness 4.

## Scope of the model

In an ideal fixed-input, lossless two-output optical setup, total detection probability is conserved, so increasing one output's probability decreases the other's. This game builds the two relative amplitudes independently and then normalizes their squared magnitudes. It does not simulate that apparatus. Real waves can have any phase; the cards use four.

## Facilitation and discussion

Start with one cancellation example, then let learners play. Ask: When is strength 1 better than strength 2? Why can adding a wave make a detector darker? Why can a good strategy lose one trial? Show the achieved chances alongside the sampled result. In Wave Lab, compare partial cancellation, complete cancellation, and a phase reversal. A simple challenge is to reach a dark detector using a strength-2 wave and smaller opposite waves.

## Credits

Developed using QCaMP material. The arrow addition model is related to the phasor method described in Feynman's QED (1985). The wave board and card game are classroom teaching models.